SECTION B (2013/2014)

STRUCTURED QUESTIONS (50 MARKS)

(**INSTRUCTION**: Please answer all questions in the space provided.)

QUESTION 1 [23 marks]

(a) Given the following Resource-Allocation Graph (RAG) as in Figure 1, identify set for *edges, E*. [3 marks]

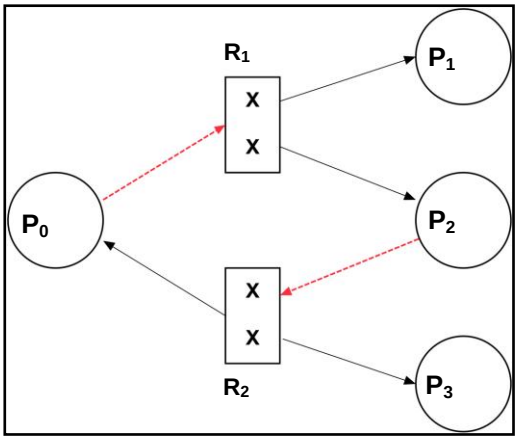


Figure 1: Resource-Allocation Graph

(b) State 2 methods for deadlock recovery. [2 marks]

(c) Consider a system with 4 processes P_0 through P_3, 3 resource types A, B, and C, and the current state at t_0 as shown in Table 1. *Hint: Safety algorithm and Resource-request algorithm in Appendix 1.*

- Resource type A has 11 instances
- Resource type B has 8 instances
- Resource type C has 6 instances

Table 1: Process allocation

Process	Allocation			Max		
	A	**B**	**C**	**A**	**B**	**C**
P_0	2	0	0	4	2	1
P_1	3	1	2	5	3	5
P_2	2	1	1	2	3	2
P_3	1	3	1	1	4	5

i) Calculate the *Need* parameter for each process based on Table 1. [2 marks]

ii) Execute the safety algorithm to determine either the system is in safe or unsafe state and write the safe sequence (if safe state). Show all your workings.

[6 marks]

iii) Assume P_1 request 1 additional resource instance of resource type B and 1 instance of resource type C. Using *resource-request algorithm* and *safety algorithm*, identify if P_1 requests can be fulfilled by determining the new safe sequence. Show all your workings.

[6 marks]

iv) After P_1's request is fulfilled in question (iii), P_3 makes the following request, $Request_3$ = (0, 0, 1). Can the system grant the request of P_3 using resource-request algorithm and safety algorithm? Explain your answer. [4 marks]

QUESTION 2 [20 marks]

(a) In an operating system, dynamic memory partition is implemented.

i) Show how the following set of jobs {Job1: 200 KB, Job2: 280 KB, Job3: 417 KB, and Job4: 500 KB} are allocated inside the main memory. [3 marks]

ii) Referring to Table 2, assume at t_4 Job3 ends and at t_6 Job1 ends producing the following new arrival sequence:

Table 2: Time and job description

Time	Job #	Job Size in KB
t_5	5	200
t_7	6	500

a) Show/draw how these new jobs are allocated into memory space at t_4 till t_7. [4 marks]

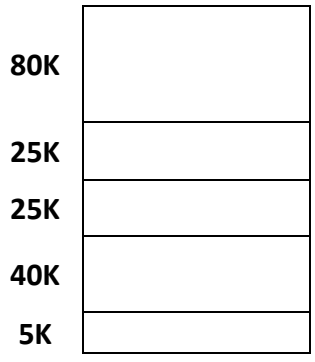
b) Is there any job that cannot be allocated into the memory? If yes, state which job and give a reason. [2 marks]

(b) Given new Jobs in Table 3 and main memory using relocatable dynamic partition in Table 4.

Table 3: Job description

Jobs	Size
1	40K
2	40K
3	20K
4	25K
5	10

Table 4: Memory partition



i) Draw and label each job that will be allocated into the memory and shades all unused partition clearly. [4 marks]

ii) Calculate the total of fragmentation of the memory for the above case. [2 marks]

iii) If another Job 6 with size 35K arrives, can this job be allocated into the memory? Suggest your solution. [5 marks]

QUESTION 3 [8 marks]

(a) Given a paging system with 2^{24} bytes of physical memory; 2^{10} pages of logical address space and page size of 512 bytes.

i) What is the total size of the logical address space. [1 mark]

ii) What is the frame size of the physical memory. [1 mark]

iii) How many frames of the physical memory. [1 mark]

iv) How many number of entry in the page table. [1 mark]

v) Calculate the offset bits for a page of virtual memory. [1 mark]

vi) Draw the frame offset of the physical memory address and page offset of logical memory. [3 marks]

QUESTION 4 [15 marks]

(a) Given the following reference string:

`1 3 2 1 4 3 2 4 1 5 2 4`

For each of the following page replacement algorithms, calculate how many **page faults** will occur if **3 frames** are available:

i) Least Recently Used (LRU) [4 marks]

ii) Optimal Page Replacement (OPT) [4 marks]

iii) Based on your answer in a i) and a ii), which algorithm is better? Justify your answer. [2 marks]

(b) Define what is thrashing**?** [1 mark]

(c) State 2 techniques to prevent thrashing. [2 marks]

(d) Based on Figure 2, explain the upper and lower bound of page fault. [2 marks]

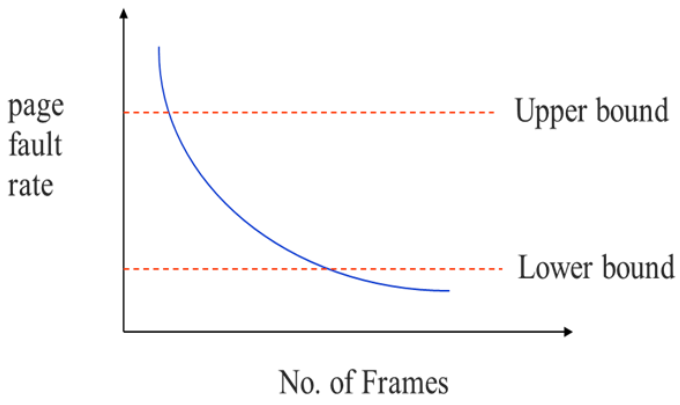


Figure 2: No. of frames vs page faulty rate

QUESTION 5 [24 Marks]

(a) Suppose a user of a UNIX system is currently in the home directory `/home/user1`. If the user issue the command `cd classes/scsr2043` and then the command `pwd`, what will be the user's current directory? Assume all directories exist. [1 mark]

(b) A user issues the Linux command `ls -l` and the following output is displayed on the screen:

```
-rw-r--r-- 1  muhalim  muhalim      0  Dec 30  13:12   fork2.c
drwxr-xr-x 2  muhalim  muhalim   4096  Dec 30  11:52   ResultC
```

i) Is `ResultC` a directory or a file? [1 mark]

ii) What is the access right of the owner of `fork2.c` [1 mark]

ii) Give a Linux command that will enable a group of users that share `fork2.c` to write to the file. [1 mark]

(c) Match the list of terms given below to each of the descriptions: [2 marks]

Access Control List	Access Control Matrix	Two-level directory
Contiguous Allocation	Indexed Allocation	Bit Map / Bit Vector
C-SCAN	SSTF	RAID	Data Stripping

i) A technique for spreading data over multiple disks.

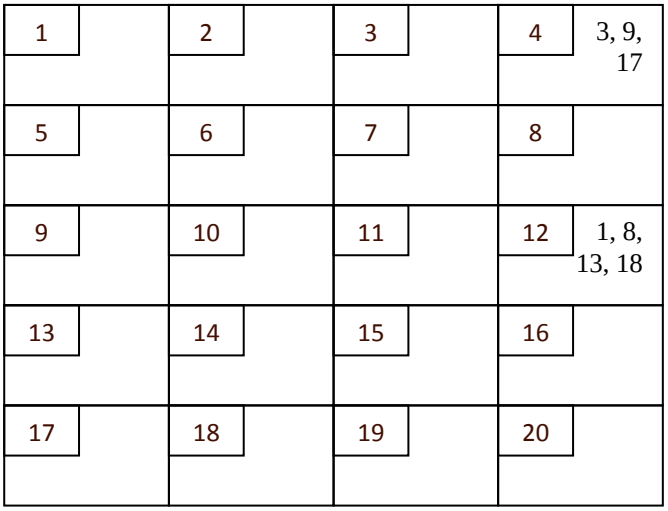
ii) A file protection mechanism that uses much more storage space as files and users increase.

iii) Provides easy direct access to records in a file (Every parts of a file are stored in same compact area).

iv) A technique to manage free spaces in disk.

(d) List 3 factors which determine how long it takes to read or write a disk block. Provide the formula to compute Access time. [2 marks]

(e) Figure 3 shows a diagram of disk storage containing two files, `File1` and `File2`:



File	Index block
File1	12
File2	4

Figure 3: Disk storage

i. Add arrows in the diagram if `File1` and `File2` start at block 12 and block 4 respectively. [3 marks]

ii. State the name of disk allocation technique that is being used. [1 mark]

(f) Figure 4 shows a diagram of disk storage containing two files, `File1` and `File2`:

Directory

File	Address	Size	Next
	1	512	
File1 (1)	2	512	8
File2 (1)	3	512	6
	. .	. .	
File2 (2)	6	512	11
	7	512	
File1 (2)	8	512	20
	. .	. .	
File2 (3)	11	512	-
	. .	. .	
File1 (3)	20	512	-

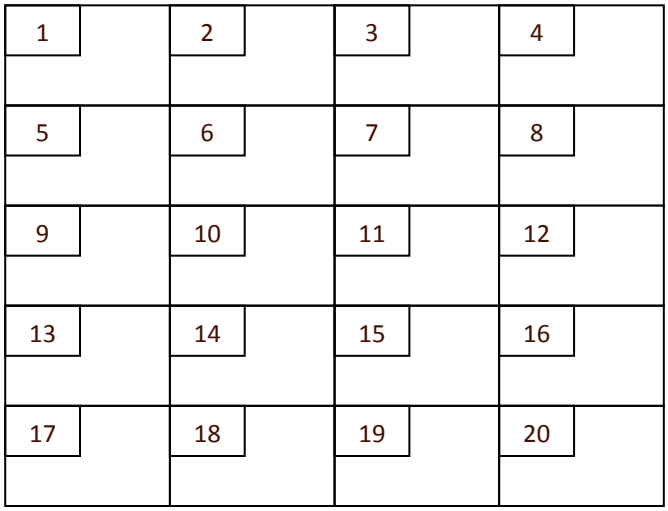


Figure 4: Disk Storage

i) Add arrows in the diagram according to the directory information for `File1` and `File2`. [3 marks]

ii) State the name of disk allocation technique that is being used. [1 mark]

(g) Suppose disk storage has block size of 4KB and disk size of 2 GB. Calculate total space needed for the bit vector. [2 marks]

(h) Assume the head of a disk is receiving track requests waiting in the queue in the following order:

98, 183, 65, 14, 122, 100, 20, 35

If the head starts at track 70 and going towards inner direction, show the movement path of the read-write head arm if the following seek strategies used:

i) C-SCAN [3 marks]

0 14 20 35 65 98 100 122 183 199

---

ii) C-LOOK [3 marks]

0 14 20 35 65 98 100 122 183 199

------- End of Question -------

GOOD LUCK !!

APPENDIX 1

Safety Algorithm

1. Let ***Work*** and ***Finish*** be vectors of length ***m*** and ***n***, respectively.
   Initialize:
   Work = Available
   Finish[i] = false for ***i***= 1, 2, …, ***n***
2. Find an index ***i*** such that both
   Finish[i] == false
   $Need_i \leq Work$
   If no such ***i*** exists, go step 4.
3. $Work = Work + Allocation_i$
   Finish[i] == true
   Go to step 2.
4. If ***Finish[i] == true*** for all ***i***→ then the system in safe state.

Resource-Request Algorithm

1. If $Request_i \leq Need_i$, go to step 2.
   Otherwise, raise an error condition since the process has exceeded its maximum claim.

2. If $Request_i \leq Available$, go to step 3.
   Otherwise, P_i must wait since the resources are not available.

3. Have the system pretend to have allocated the requested resources to process P_i by modifying the state as follows:

   $Available := Available - Request_i$
   $Allocation_i := Allocation_i + Request_i$
   $Need_i := Need_i - Request_i$

4. Execute the **safety algorithm** (Figure B.5) on a **new state**. If the resulting state is safe, allocate the resources, else delay the request.

SECTION B (2014/2015)

STRUCTURED QUESTIONS (80 MARKS)

(**INSTRUCTION**: Please answer all questions in the space provided.)

QUESTION 1 (20 MARKS)

a. You are given the following set of edges. Use it to answer the following questions.

Edge, E = {P0 → R1, P1 → R1, R1→P2, P2→R2, R2→P1, R3→P0}

i. Draw the Resource Allocation Graph. [3]

ii. Using the RAG in (i) for the scenarios given below, please explain the possibility of deadlock happening (if deadlock happens). [4]

Scenario A

R1 has 2 instances, R2 has 2 instances, R3 has 1 instance

Scenario B

R1 has 1 instance, R2 has 2 instances, R3 has 2 instances

b. Consider the following system has three processes, P0, P1 and P2, and with three resource types, A, B and C. At time t_0, the resource allocation state of the system is as shown below:

	Max			**Allocation**		
	A	**B**	**C**	**A**	**B**	**C**
P0	4	2	3	2	1	1
P1	3	1	6	1	0	2
P2	6	3	3	3	2	2

Available		
A	**B**	**C**
2	1	2

i. Calculate the total number of resource for type A, B and C. Show your calculation. [2]

ii. What is the content of the Need matrix ? [2]

iii. Determine whether the system is in **safe** or **unsafe state** at t_0. [5]

iv. If at t_1, P2 request for (1,1,1), should the system satisfy the request of P2? Show your works. [4]

QUESTION 2 (22 MARKS)

a. In an operating system, assume memory is allocated as specified in Figure 1.

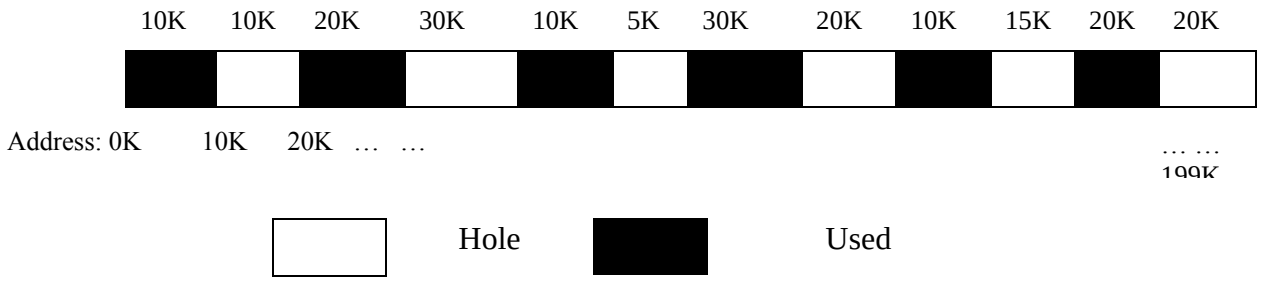


Figure 1: Variable partition memory allocation.

i. Show how the following set of jobs {J1 - 20KB, J2 - 10KB, J3 - 5KB} (in that order) are allocated inside the main memory using **first-fit allocation**. At what starting address will each job be allocated? [3]

ii. Show how the following set of jobs {J1 - 20KB, J2 - 10KB, J3 - 5KB} (in that order) are allocated inside the main memory using **best-fit allocation**. At what starting address will each job be allocated? [3]

b. Given new jobs Job 6 until Job 10 in Figure 2 and main memory using *relocatable dynamic partition* in Figure 3. Consider Job 1 until 5 end at t_{10}.

Jobs	Size
6	40K
7	40K
8	20K
9	25K
10	10K

Figure 2: Job description

80K	Job 1
25K	Job 2
25K	Job 3
40K	Job 4
5K	Job 5

Figure 3: Memory partition

i. Redraw and label each job that will be allocated into the memory after t_{10}. Shades all unused partition clearly. [4]

ii. Calculate the total fragmentation of the memory for case (i). [2]

iii. If another Job 11 with size 35K arrives, can this job be allocated into the memory? Suggest your solution. [1]

c. What are the **TWO** major differences between paging and segmentation? [2]

d. Given a paging system with 2^{24} bytes of physical memory; 2^{10} pages of logical address space and page size of 512 bytes.

i. What is the total size of the logical address space. [1]
ii. What is the frame size of the physical memory. [1]
iii. How many frames of the physical memory. [1]
iv. How many number of entry in the page table. [1]
v. Calculate the offset bits for a page of virtual memory. [1]
vi. Draw the frame offset of the physical memory address and page offset of logical memory. [2]

QUESTION 3 (15 MARKS)

a. A page replacement algorithm should minimize the number of page faults. Consider the following page reference string:

1, 2, 3, 4, 5, 3, 4, 1, 6, 7, 8, 7, 8, 9, 7, 8, 9, 5, 4, 5, 4, 2

Assuming demand paging with **FOUR** page frames, how many page faults would occur for the following replacement algorithms? Show all your works.

i. FIFO [3]

ii. OPT [4]

iii. Compare and discuss both algorithms in term of page faults occurred? Which do you think is better? [2]

b. In paging, thrashing may occur.

i. What is the cause of thrashing? [2]

ii. How does the system detect thrashing? [2]

iii. Once it detects thrashing, what can the system do to eliminate this problem? [2]

QUESTION 4 (23 MARKS)

a. Indicate whether statements below are **TRUE** or **FALSE** statements, according to the Unix files access control as shown in Figure 4.

```
-r-xr--r-x  Ali   Ali        2.6K  Feb 16 2011      Untitled
-rw-r-xr--  Abu   Student    436B  Aug 28 2011      README
drwxr-xr-x  Ali   Student    197B  Aug 28 2011      Notes
```

Figure 4: Unix file access control

i. Only Untitled and README are files. [1]
ii. Notes is a folder that is shared among users of the system. [1]
iii. Owner of the README file is Student. [1]
iv. Untitled file is editable by Ali. [1]
v. README is only editable by Abu. [1]

b. Assuming Figure 5 shows a storage map of a disk with capacity of 400KB. The disk has 20 storage blocks and size of a storage block is 20KB. Two files will be stored in the storage disk that are **File1** and **File2**. Size of File1 is **58KB**, while the size of File2 is **36KB**. Assuming the operating system is using **Indexed Storage Allocation**. Each storage block can only store 3 storage block addresses.

i. Fill in the Index Block Number column in Directory File, and update the Storage Map accordingly for File1. Consider Index block for File 1 is **block 3**. [2]
ii. Fill in the Index Block Number column in Directory File, and update the Storage Map accordingly for File2. Consider Index block for File 2 is **block 18**. [2]

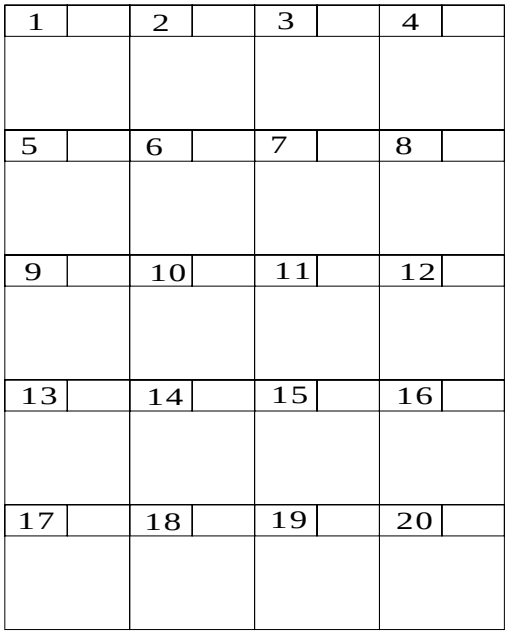


Figure 5: Storage Map

Filename	File Size (KB)	Index Block Number
File1	58	
File2	36	

Directory File

c. Figure 6 shows a bit map for free space management for a hypothetical Operating System whereby bit **0** indicates that the block is used and bit **1** indicates that the block is **free**.

Block No	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
Bit Map	0	1	1	0	1	0	0	1	0	1	1	1	0	1	0	0

Figure 6: Storage Map

Suppose that the size of a block is 15KB.

i. List the block numbers that are allocated to a file named DATA.txt. The size of the DATA.txt is 64KB. [3]

ii. Is there any block that suffers from fragmentation in storing the DATA.txt? If your answer is a YES, state the block number that contains the fragmentation. [2]

d. On a disk with 900 tracks: numbered 0 to 899. The disk queue in FIFO order contains requests on the following tracks:

142, 778, 292, 575, 205, 376

Starting from the current head location **545**, sketch figure to illustrate the head movement (track traveled) to serve all the requests based on

i. **SSTF** seek strategy. [3]

ii. **C-LOOK** seek strategy. The head movement direction is outer to inner. [3]

iii. According to the above scenario (i) and (ii), which seek strategy is better between SSTF and C-LOOK? Briefly justify your answer. [3]

SECTION B (2015/2016)

STRUCTURED QUESTIONS (80 MARKS)

(**INSTRUCTION**: Please answer all questions in the answer booklet)

QUESTION 1 [25 MARKS]

a). Answer the following questions based on the basic concepts of memory management for operating systems.

i. The management of main memory is critical, thus the entire system performance is dependent on two (2) items/factors. Give the **two (2)** items/factors. [2]

ii. Describe the fundamental difference between **logical address** and **physical address**? [2]

iii. A process can be swapped temporarily out of memory to a backing store and then brought back into memory for continued execution. In the simplest term, what does **backing store** here refers to? [2]

iv. When **a process is being swap out** and another desired **process is being swap in by the dispatcher** for execution, what does it actually means? [2]

v. Swapping to free up memory is constrained by some factors. **Can a process waiting for I/O operation be swapped?** Furthermore, what is the **primary selection criterion to swap out** a process? [2]

b). The following questions deals specifically with the **fixed partitions memory scheme**.

i. As shown in figure below, there are **nine (9) fixed partitions of different sizes,** and there are several jobs waiting to be loaded into memory for execution. How would the **First-fit**, **Best-fit** and **Worst-fit** algorithms place all the jobs into the memory partitions? [8]

Memory Partitions

PARTITION NO.	PARTITION SIZE(KB)
1	50
2	110
3	80
4	150
5	100
6	90
7	130
8	160
9	60

Jobs List

JOBS	SIZE OF JOB (KB)
A	60
B	120
C	40
D	30
E	80
Z	100

Figure: Jobs Allocation into Fixed Partitions

Allocation Strategy / Partition	1	2	3	4	5	6	7	8	9
Next-fit (This has been done for you)		**A**		**B**	**C**	**D**	**E**	**Z**	
First-fit									
Best-fit									
Worst-fit									

ii. Calculate the **size of internal fragmentations incurred for each job** and then compute the **total size of fragmentation for First-fit and Best-fit** job allocation schemes. [4]

JOBS	First-fit internal fragmentation	Best-fit internal fragmentation
A		
B		
C		
D		
E		
Z		
Total size of internal fragmentation:		

iii. For all the jobs allocation schemes above, **is there any job that needs to wait from entering a partition?** State the job and allocation scheme. [1]

iv. **Rank all the three (3) jobs allocation** schemes above (First-fit, Best-fit, Worst-fit) in terms of how efficiently they use memory. [2]

Rank 1:

Rank 2:

Rank 3:

QUESTION 2 [10 MARKS]

a). State **TRUE** or **FALSE** for the following statements. [3]

Statements	TRUE/ =FALSE
Paged memory allocation divides each incoming job into pages of equal size.	
The disadvantage of paged memory is that it allows job allocation in non contiguous memory.	
Page table consist of page number and corresponding page frame memory address.	

b). Given a paging system with 2^{32} bytes of physical memory; **2^{10} pages** of logical address space and **page size of 4096 bytes**. [7]

i. What is the total size of the logical address space.

ii. What is the frame size of the physical memory.

iii. How many frames of the physical memory.

iv. How many number of entry in the page table.

v. Calculate the offset bits for a page of virtual memory.

vi. Draw the frame offset of the physical memory address and page offset of logical memory.

QUESTION 3 [15 MARKS]

CPU is executing a loop for an array implementation and a series of memory locations are accessed repeatedly. Given the following **reference string**:

1 2 3 4 1 2 3 4 1 2

a). Apply **LRU page replacement** algorithm and **calculate how many page faults** will occur. Assume **three (3) page frames** are used. **Show all your workings.** [4]

b). Apply **OPT page replacement** algorithm and **calculate how many page faults** will occur. Assume **three (3) page frames** are used. **Show all your workings.** [4]

c). Which algorithm has the **best performance in term of page faults occurred**? **Justify your answer.** [1]

d). **State 2 methods** used in LRU implementation. [2]

e). Does increasing the **page frames to four (4) cause Belady's anomaly** effect on LRU? **Justify your answer.** [4]

QUESTION 4 [10 MARKS]

a). Arrange READ logical command statements in correct sequence when interacting with file manager. [2]

Statements	Sequence
Wait for rotational delay (sector containing record passes under read/write head)	
Send flag indicating free device for another request	
Activate appropriate read/write head and read record	
Transfer record to main memory	
Move read/write heads to record cylinder	**1**

b). Indicate whether statements below are **TRUE** or **FALSE** according to the Unix files access control list as shown in figure below. [6]

```
-rwxr--r-x   Ali    Ali          2.6K      Feb  16  2011      Untitled
-rw-r-xr-x   Abu   Student     436B   Aug  28  2011    README
drwxrwxr-x   Ali   Student     197B   Aug  28  2011    Notes
```

Figure: Unix file access control

i. Untitled and README are directory.

ii. README group access right is read only.

iii. Notes is a folder that is shared among Student group.

iv. Owner of the README file is Student.

v. Untitled file can be edited by Ali.

vi. README can be executed by public.

c). The following figure below shows some output from file-system related commands on a local Linux system. Use this output and your knowledge of Linux file systems to answer the following questions. [2]

```
[carey@csl]$ df
```

Filesystem	1K-blocks	Used	Available	Use%	Mounted on
/dev/sda2	20315844	7513824	11753380	39%	/
/dev/sda5	10581704	5181092	4854404	52%	/tmp
/dev/sda1	1019208	50356	916244	6%	/boot
tmpfs	1684784	77164	1607620	5%	/dev/shm
nsi:/export/research	247709760	203879488	31247360	87%	/home/research
nse:/export/scratch	3361364788	528637368	2661979928	17%	/home/scratch
nsj:/export/proj/dsl	485097928	184513976	275942416	41%	/home/dsl
nsg:/export/ug	381885660	79136396	283350608	22%	/home/ugc
nsf:/export/ug	81885660	88768580	273718424	25%	/home/ugb
nsh:/export/grads	789574392	507946744	241519616	68%	/home/grads
nsb:/export/ug	381885660	76958700	285528304	22%	/home/uga

Figure X: Unix file-system output

i. How many different file systems are accessible on this Linux system?

ii. Which file system is the fullest (in terms of percent occupancy)?

iii. Which file system has the largest physical storage capacity?

iv. Which file system has the fewest bytes currently stored?

QUESTION 5 [10 MARKS]

Figure below shows a storage map of a disk. The disk has **twenty (20) storage blocks and size of a storage block is 256KB**. Two files will be stored in the storage disk are **DATA1** and **DATA2**. Size of DATA1 is **718KB**, while the size of DATA2 is **300KB**.

Assuming the operating system is using **Indexed Storage Allocation**, each storage block can only **store 4 block addresses**.

1		2		3		4	
5		6		7		8	
9		10		11		12	
13		14		15		16	
17		18		19		20	

Figure Storage Map

Filename	File Size (KB)	Index Block Number
DATA1	718	
DATA2	300	

Directory File

a). What is the **maximum capacity of the disk**? [2]

b). Fill in the Index Block Number column in Directory File and update the Storage Map accordingly for DATA1. Consider index block for DATA1 is **block 6** and use blocks with the **lowest number value** for DATA1 storage map. [3]

c). Fill in the Index Block Number column in Directory File and update the Storage Map accordingly for `DATA2`. Consider Index block for `DATA2` is **block 16** and use blocks with the **highest number value** for `DATA2` storage map. [3]

d). **Is there any block** that suffers **from fragmentation in storing the `DATA2` file**? If your answer is a YES, **state the block number** and **calculate the size of fragmentation** that incurred in the block. [2]

QUESTION 6 [10 MARKS]

On a disk with 800 tracks: **numbered from 0 to 799**. The disk queue in FIFO contains the following track requests:

233, 75, 744, 300, 680, 190

Starting from the current head location **600**, sketch figure to illustrate the head movement (track traveled) to serve all the requests based on:

a). **SSTF** seek strategy. [4]

0 75 190 233 300 600 680 744 799

b). **C-SCAN** seek strategy. The head movement direction is **outer to inner**. [4]

0 75 190 233 300 600 680 744 799

c). According to the above scenario in **(a) and (b)**, which seek **strategy is better** between **SSTF** and **C-SCAN**? **Briefly justify your answer**. [2]